

Sec. 15.1

練習作業 Homework : 6,14,16,19

繳交作業 Submission :

6 : 計算逐次積分。

$$I = \int_0^3 \int_{-2}^0 (x^2 y - 2xy) dy dx \quad \text{先對 } y \text{ 積分，視 } x \text{ 為常數}$$

$$\begin{aligned} \int (x^2 y - 2xy) dy &= (x^2 - 2x) \int y dy \quad \text{含 } x \text{ 的項為常數，提出} \\ &= (x^2 - 2x) \left(\frac{1}{2} y^2 \right) \end{aligned}$$

$$I = \int_0^3 (x^2 - 2x) \left(\frac{1}{2} y^2 \right)_{-2}^0 dx \quad \left(\frac{1}{2} y^2 \right)_{-2}^0 = 0 - 2 = -2$$

$$= \int_0^3 (x^2 - 2x)(-2) dx \quad \text{這時不會有含 } y \text{ 的項}$$

$$= -2 \left(\frac{1}{3} x^3 - x^2 \right) \Big|_0^3 = -2[(9 - 9) - (0 - 0)] = 0$$

14 : 計算二重積分 $\iint_R \frac{\sqrt{x}}{y^2} dA$, $R: 0 \leq x \leq 4, 1 \leq y \leq 2$ 。

Hint : $\int \sqrt{x} dx$, $\int \frac{1}{y^2} dy$, which one is more easy? 皆可

$$I = \int_1^2 \int_0^4 \frac{\sqrt{x}}{y^2} dx dy \quad \text{先對 } x \text{ 積分，視 } y \text{ 為常數}$$

$$= \int_1^2 \frac{1}{y^2} \left(\int_0^4 x^{1/2} dx \right) dy \quad \text{含 } y \text{ 的項為常數，提出}$$

$$= \int_1^2 \frac{1}{y^2} \left(\frac{2}{3} x^{3/2} \right)_0^4 dy \quad \left(\frac{2}{3} x^{3/2} \right)_0^4 = \frac{16}{3} - 0 = \frac{16}{3}$$

$$= \frac{16}{3} \int_1^2 \frac{1}{y^2} dy$$

$$= \frac{16}{3} \left(-\frac{1}{y} \right)_1^2 = \frac{16}{3} \left[-\frac{1}{2} - (-1) \right] = \left(\frac{16}{3} \right) \left(\frac{1}{2} \right) = \frac{8}{3}$$

14：計算二重積分 $\iint_R \frac{\sqrt{x}}{y^2} dA$ ， $R: 0 \leq x \leq 4, 1 \leq y \leq 2$ 。

$$I = \int_0^4 \int_1^2 \frac{\sqrt{x}}{y^2} dy dx \quad \text{先對 } y \text{ 積分，視 } x \text{ 為常數}$$

$$= \int_0^4 \sqrt{x} \left(\int_1^2 \frac{1}{y^2} dy \right) dx \quad \text{含 } x \text{ 的項為常數，提出}$$

$$= \int_0^4 \sqrt{x} \left(\frac{-1}{y} \Big|_1^2 \right) dx \quad \frac{-1}{y} \Big|_1^2 = \frac{-1}{2} - (-1) = \frac{1}{2}$$

$$= \frac{1}{2} \int_0^4 \sqrt{x} dx = \frac{1}{2} \left(\frac{2}{3} x^{3/2} \right) \Big|_0^4 = \frac{1}{3} (8 - 0) = \frac{8}{3}$$

16：計算二重積分 $\iint_R y \sin(x+y) dA$ ， $R: -\pi \leq x \leq 0, 0 \leq y \leq \pi$ 。

Hint： x, y 積分上下限均為常數， $\int y \sin(x+y) dx$ ， $\int y \sin(x+y) dy$ 那個較為容易？

完全看被積分函數，瞭解？

$\int y \sin(x+y) dy$ 積分技巧如同 $\int y \sin(1+y) dy$ integration by parts

$\int y \sin(x+y) dx$ 積分技巧如同 $2 \int \sin(x+2) dx$ 這個較為單純

$$I = \int_0^\pi \int_{-\pi}^0 y \sin(x+y) dx dy \quad \text{先對 } x \text{ 積分，視 } y \text{ 為常數}$$

$$= \int_0^\pi y \left(\int_{-\pi}^0 \sin(x+y) dx \right) dy \quad \text{含 } y \text{ 的項為常數，提出}$$

$$= \int_0^\pi y [-\cos(x+y)]_{-\pi}^0 dy \quad [-\cos(x+y)]_{-\pi}^0 = (-\cos y) - [-\cos(y-\pi)] = -2 \cos y$$

$$\because \cos(y-\pi) = \cos y \cos \pi + \sin y \sin \pi = -\cos y$$

$$= \int_0^\pi -2y \cos y dy \quad \text{integration by parts}$$

$$= -2[y \sin y + \cos y]_0^\pi$$

$$= -2[(0-1) - (0+1)] = (-2)(-2) = 4$$

19：計算二重積分 $\iint_R \frac{xy^3}{x^2+1} dA$ ， $R: 0 \leq x \leq 1, 0 \leq y \leq 2$ 。

Hint： x, y 積分上下限均為常數， $\int \frac{xy^3}{x^2+1} dx$ ， $\int \frac{xy^3}{x^2+1} dy$ 那個較為容易？看被積分函數

$$\int \frac{xy^3}{x^2+1} dx = y^3 \int \frac{x}{x^2+1} dx, \quad \int \frac{xy^3}{x^2+1} dy = \frac{x}{x^2+1} \int y^3 dy,$$

$\int y^3 dy$ 比 $\int \frac{x}{x^2+1} dx$ 容易， $dydx$ 積分順序較佳

$$I = \int_0^1 \int_0^2 \frac{xy^3}{x^2+1} dy dx \quad \text{先對 } y \text{ 積分，視 } x \text{ 為常數}$$

$$= \int_0^1 \frac{x}{x^2+1} \left(\int_0^2 y^3 dy \right) dx \quad \text{含 } x \text{ 的項為常數，提出}$$

$$= \int_0^1 \frac{x}{x^2+1} \left(\frac{1}{4} y^4 \Big|_0^2 \right) dx \quad \frac{1}{4} y^4 \Big|_0^2 = \frac{1}{4} (16 - 0) = 4$$

$$= \int_0^1 \frac{4x}{x^2+1} dx \quad \text{代換積分法}$$

$$= 2 \ln(x^2+1) \Big|_0^1 = 2 \ln 2 - 2 \ln 1 = 2 \ln 2$$

[代換積分法，

$$u = x^2 + 1 \Rightarrow \frac{du}{dx} = 2x \Rightarrow du = 2x dx, \quad \int \frac{4x}{x^2+1} dx = 2 \int \frac{1}{u} du = 2 \ln|u| + C = 2 \ln(x^2+1) + C]$$

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Sec. 15.2

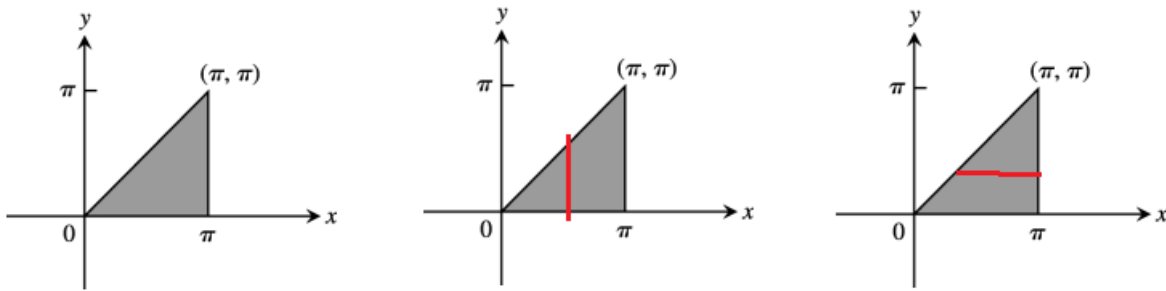
練習作業 Homework : 19,41,56,51

19. (modified)

Evaluate $I = \iint_R x \sin y dA$, where R is the region bounded by the lines

$y = x$, $y = 0$, and $x = \pi$.

繪出積分區域



只有一塊積分區域，積分順序 $dydx$ 或 $dx dy$ 皆可，採用 $dydx$

Find the intersection of $y = x$, and $x = \pi$, $\Rightarrow x = \pi, y = \pi$ 交點

$$\begin{aligned}
 I &= \iint_R x \sin y dA \\
 &= \int_0^\pi \int_0^x x \sin y dy dx = \int_0^\pi x (-\cos y)_0^x dx = \int_0^\pi x(1 - \cos x) dx \\
 &= \int_0^\pi x dx - \int_0^\pi x \cos x dx \quad \text{integration by parts} \\
 &= \frac{1}{2}(x^2)_0^\pi - (x \sin x + \cos x)_0^\pi = \frac{1}{2}\pi^2 - [(0-1) - (0+1)] = \frac{\pi^2}{2} + 2
 \end{aligned}$$

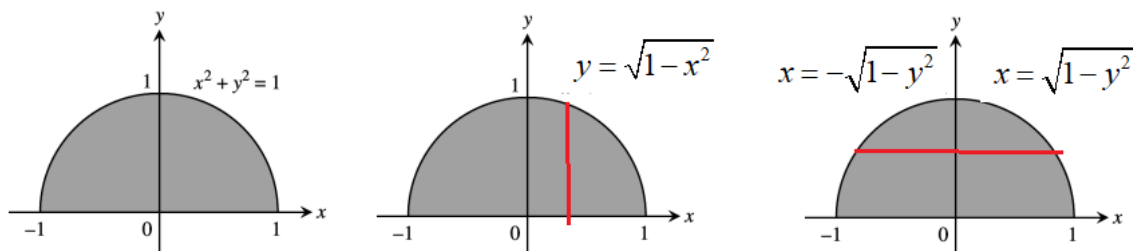
若採用 $dx dy$

$$\begin{aligned}
 I &= \iint_R x \sin y dA \\
 &= \int_0^\pi \int_y^\pi (\sin y) x dx dy = \int_0^\pi \sin y \left(\frac{1}{2} x^2 \right)_y^\pi dy \\
 &= \frac{1}{2} \pi^2 \int_0^\pi \sin y dy - \frac{1}{2} \int_0^\pi y^2 \sin y dy \\
 &= \frac{1}{2} \pi^2 [-\cos y]_0^\pi - \frac{1}{2} [-y^2 \cos y + 2y \sin y + 2 \cos y]_0^\pi \\
 &= \frac{1}{2} \pi^2 [1-1] - \frac{1}{2} [(\pi^2 + 0 - 2) - (0 + 0 + 2)] = \frac{\pi^2}{2} - 2
 \end{aligned}$$

41. (modified)

Evaluate $I = \iint_R 3y dA$, where R is the region bounded by the lines $y=0$ and the upper unit circle $x^2 + y^2 = 1$.

繪出積分區域



只有一塊積分區域，積分順序 $dydx$ 或 $dxdy$ 皆可，採用 $dydx$

$$\begin{aligned} I &= \iint_R 3y dA \\ &= \int_{-1}^1 \int_0^{\sqrt{1-x^2}} 3y dy dx = \int_{-1}^1 \frac{3}{2} (y^2)_0^{\sqrt{1-x^2}} dx = \int_{-1}^1 \frac{3}{2} (1-x^2) dx \\ &= \left(\frac{3}{2}x - \frac{1}{2}x^3 \right)_{-1}^1 = \left(\frac{3}{2} - \frac{1}{2} \right) - \left(-\frac{3}{2} + \frac{1}{2} \right) = 1 + 1 = 2 \end{aligned}$$

若採用 $dxdy$

$$\begin{aligned} I &= \iint_R 3y dA \\ &= \int_0^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} 3y dx dy = \int_0^1 3y(x)_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} dy = \int_0^1 6y\sqrt{1-y^2} dy \quad \text{substitution} \\ &= [-2(1-y^2)^{3/2}]_0^1 = 0 - (-2) = 2 \end{aligned}$$

[代換積分法，

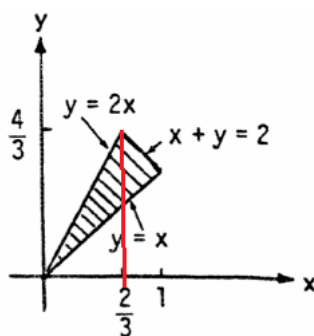
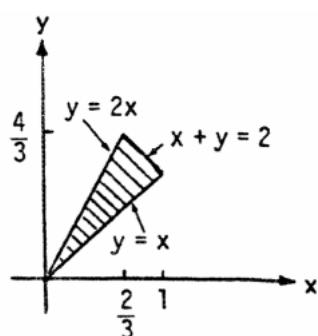
$$u = 1 - y^2 \Rightarrow \frac{du}{dy} = -2y \Rightarrow du = -2y dy,$$

$$\int 6y\sqrt{1-y^2} dy = -3 \int u^{1/2} du = -3 \left(\frac{2}{3} u^{3/2} \right) + C = -2(1-y^2)^{3/2} + C]$$

56. Evaluate $I = \iint_R xy dA$, where R is the region bounded by the lines

$y = x$, $y = 2x$, and $x + y = 2$.

繪出積分區域



積分順序 $dydx$ 或 $dx dy$ 皆可，採用 $dydx$ ，有兩塊積分區域

Find the intersection of $y = 2x$, and $x + y = 2$, $2x = 2 - x \Rightarrow x = \frac{2}{3}, y = \frac{4}{3}$ 交點

Find the intersection of $y = x$, and $x + y = 2$, $x = 2 - x \Rightarrow x = 1, y = 1$ 交點

$$\begin{aligned}
 I &= \int_0^{2/3} \int_x^{2x} xy dy dx + \int_{2/3}^1 \int_x^{2-x} xy dy dx \\
 &= \int_0^{2/3} x \left(\frac{1}{2} y^2 \right)_x^{2x} dx + \int_{2/3}^1 x \left(\frac{1}{2} y^2 \right)_x^{2-x} dx = \int_0^{2/3} \frac{3}{2} x^3 dx + \int_{2/3}^1 (2x - 2x^2) dx \\
 &= \frac{3}{8} (x^4)_0^{2/3} + (x^2 - \frac{2}{3} x^3)_1^{2/3} = \frac{6}{81} + \frac{7}{81} = \frac{13}{81}
 \end{aligned}$$

51：逐次積分 $I = \int_0^{2\sqrt{\ln 3}} \int_{y/2}^{\sqrt{\ln 3}} e^{x^2} dx dy$ ，繪出積分區域，改寫積分順序及計算積分。

Sketch the region of integration, reverse the order of integration, and evaluate the integral.

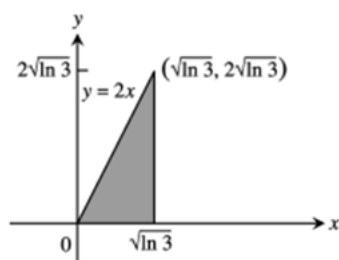
解：

$\int e^{x^2} dx$ 如何積分？不知道，只能更改積分順序

從原積分順序 $dx dy$ 與上下限知積分區域 $R: 0 \leq y \leq 2\sqrt{\ln 3}$ ， $y/2 \leq x \leq \sqrt{\ln 3}$

先畫出邊界 $x = y/2$ 及 $x = \sqrt{\ln 3}$ 圖形，再由 $0 \leq y \leq 2\sqrt{\ln 3}$ 決定積分範圍。

$$x = y/2 \Rightarrow y = 2x$$



新積分順序 $dy dx$ ，

y 值的變動從 $y = 0$ 到 $y = 2x$ ，

而 x 值左起 $x = 0$ 右至 $x = \sqrt{\ln 3}$

$$R: 0 \leq x \leq \sqrt{\ln 3}, 0 \leq y \leq 2x$$

故 $I = \int_0^{\sqrt{\ln 3}} \int_0^{2x} e^{x^2} dy dx$ ，計算如下

$$I = \int_0^{\sqrt{\ln 3}} \int_0^{2x} e^{x^2} dy dx = \int_0^{\sqrt{\ln 3}} e^{x^2} (y|_0^{2x}) dx = \int_0^{\sqrt{\ln 3}} 2xe^{x^2} dx = (e^{x^2})_0^{\sqrt{\ln 3}} = e^{\ln 3} - 1 = 3 - 1 = 2$$

(substitution, $u = x^2 \Rightarrow du = 2x dx$, $\int 2xe^{x^2} dx = \int e^u du = e^u + C = e^{x^2} + C$)

54：逐次積分 $I = \int_0^8 \int_{\sqrt[3]{x}}^2 \frac{1}{y^4+1} dy dx$ ，繪出積分區域，改寫積分順序及計算積分。

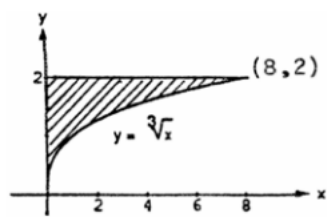
Sketch the region of integration, reverse the order of integration, and evaluate the integral.

解：

$\int \frac{1}{y^4+1} dy$ 如何積分？不知道，只能更改積分順序

從原積分順序 $dy dx$ 與上下限知積分區域 $R: \sqrt[3]{x} \leq y \leq 2, 0 \leq x \leq 8$

先畫出邊界 $y = x^{1/3}$ 或 $(x = y^3)$ 及 $y = 2$ 圖形，再由 $0 \leq x \leq 8$ 決定積分範圍。



新積分順序 $dx dy$ ，

x 值的變動從 $x = 0$ 到 $x = y^3$ ，

而 y 值下起 $y = 0$ 上至 $y = 2$

$R: 0 \leq x \leq y^3, 0 \leq y \leq 2$

故 $I = \int_0^2 \int_0^{y^3} \frac{1}{y^4+1} dx dy$ ，計算如下

$$I = \int_0^2 \int_0^{y^3} \frac{1}{y^4+1} dx dy = \int_0^2 \frac{1}{y^4+1} (x)_0^{y^3} dy = \int_0^2 \frac{y^3}{y^4+1} dy = \frac{1}{4} [\ln(y^4+1)]_0^2 = \frac{1}{4} [\ln 17 - \ln 1] = \frac{1}{4} \ln 17$$

$$(\text{substitution}, u = y^4 + 1 \Rightarrow du = 4y^3 dy, \int \frac{1}{y^4+1} dy = \frac{1}{4} \int \frac{1}{u} du = \frac{1}{4} \ln|u| + C = \frac{1}{4} \ln|y^4+1| + C)$$

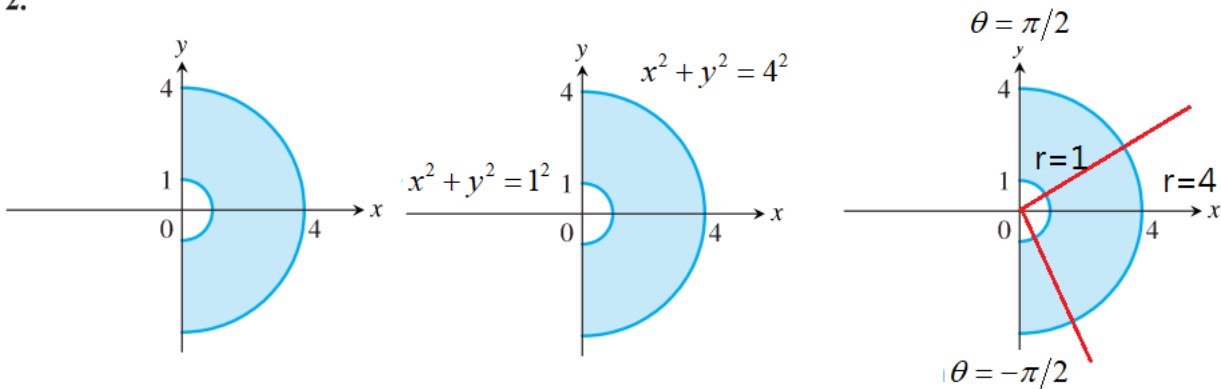
Sec.15.4

練習作業 Homework : 2,3,4,7,8,10,17,22

Regions in Polar Coordinates

In Exercises 1–8, describe the given region in polar coordinates.

2.



極座標與 Cartesian 座標: $x = r \cos \theta$, $y = r \sin \theta$, $x^2 + y^2 = r^2$

inside circle $x^2 + y^2 = 1^2 \Rightarrow r^2 = 1 \Rightarrow r = 1$, outside circle $x^2 + y^2 = 4^2 \Rightarrow r^2 = 4^2 \Rightarrow r = 4$,

Draw a ray which starts from the origin O and passes through the shadow region, the ray enters at the smaller circle $r = 1$ and leaves at the bigger circle $r = 4$.

i.e., $1 \leq r \leq 4$.

The shadow region is over the fourth and first quadrant.

The angle rotates countclockwise from $\theta = -\pi/2$ to $\theta = \pi/2$.

i.e., $-\pi/2 \leq \theta \leq \pi/2$.

Thus, $G: 1 \leq r \leq 4, -\pi/2 \leq \theta \leq \pi/2$.

內圓 $x^2 + y^2 = 1^2 \Rightarrow r^2 = 1 \Rightarrow r = 1$, 外圓 $x^2 + y^2 = 4^2 \Rightarrow r^2 = 4^2 \Rightarrow r = 4$

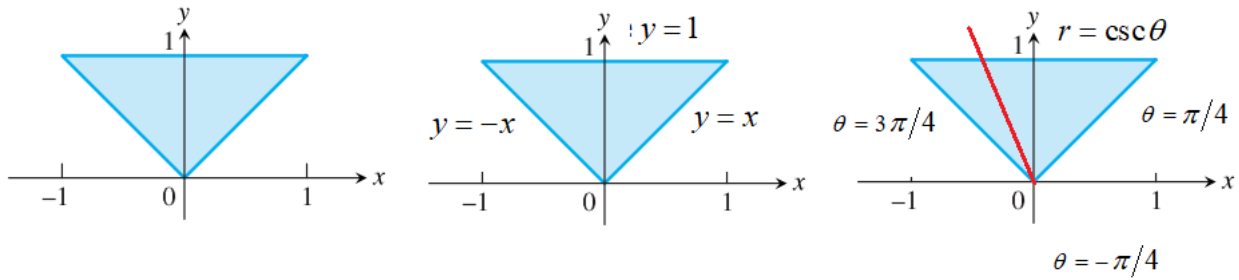
畫一條射線，從原點 O 開始，通過陰影區域，射線從小圓 $r = 1$ 進入，從大圓 $r = 2$ 離開。
即是 $1 \leq r \leq 4$ 。

陰影區域落在第 4 及第 1 象限，

角度從 $\theta = -\pi/2$ 開始逆時針旋轉到 $\theta = \pi/2$ 。即是 $-\pi/2 \leq \theta \leq \pi/2$ 。

因此， $G: 1 \leq r \leq 4, -\pi/2 \leq \theta \leq \pi/2$ 。

3.



極座標與 Cartesian 座標: $x = r \cos \theta$, $y = r \sin \theta$, $x^2 + y^2 = r^2$

$$\text{Left line } y = -x \Rightarrow r \sin \theta = -r \cos \theta \Rightarrow \frac{\sin \theta}{\cos \theta} = -1 \Rightarrow \tan \theta = -1$$

$$\Rightarrow \theta = -\frac{\pi}{4} \text{ in the fourth quadrant or } \theta = \frac{3\pi}{4} \text{ in the second quadrant.}$$

$$\text{take } \theta = \frac{3\pi}{4}, \text{ since } \theta \text{ in the second quadrant.}$$

$$\text{Right line } y = x \Rightarrow r \sin \theta = r \cos \theta \Rightarrow \frac{\sin \theta}{\cos \theta} = 1 \Rightarrow \tan \theta = 1 \Rightarrow \theta = \frac{\pi}{4}$$

$$\text{The angle rotates counterclockwise from } \theta = \frac{\pi}{4} \text{ to } \theta = \frac{3\pi}{4} . \text{ i.e., } \frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4} .$$

$$\text{Horizontal line } y = 1 \Rightarrow r \sin \theta = 1 \Rightarrow r = \frac{1}{\sin \theta} \Rightarrow r = \csc \theta$$

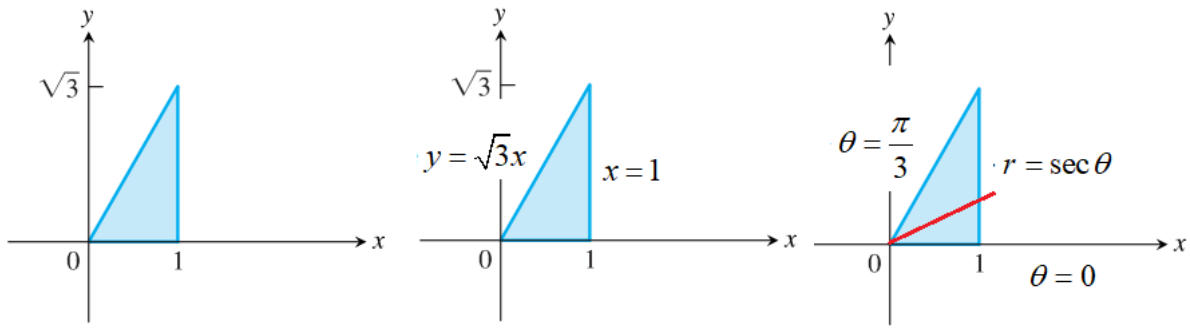
Draw a ray which starts from the origin O and passes through the shadow region.

The ray enters at the origin O, $r = 0$ and leaves at the horizontal line $r = \csc \theta$.

$$\text{i.e., } 0 \leq r \leq \csc \theta .$$

$$\text{Thus, } G : 0 \leq r \leq \csc \theta , \frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4} .$$

4.



極座標與 Cartesian 座標: $x = r \cos \theta$, $y = r \sin \theta$, $x^2 + y^2 = r^2$

Left line $y = \sqrt{3}x \Rightarrow r \sin \theta = \sqrt{3}r \cos \theta \Rightarrow \frac{\sin \theta}{\cos \theta} = \sqrt{3} \Rightarrow \tan \theta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3}$.

The angle rotates counterclockwise , in the first quadrant ,

from $\theta = 0$ to $\theta = \frac{\pi}{3}$. i.e., $0 \leq \theta \leq \frac{\pi}{3}$.

Right vertical line $x = 1 \Rightarrow r \cos \theta = 1 \Rightarrow r = \frac{1}{\cos \theta} \Rightarrow r = \sec \theta$

Draw a ray which starts from the origin O and passes through the shadow region.

The ray enters at the origin O, $r = 0$ and leaves at the vertical line $r = \sec \theta$.

i.e., $0 \leq r \leq \sec \theta$.

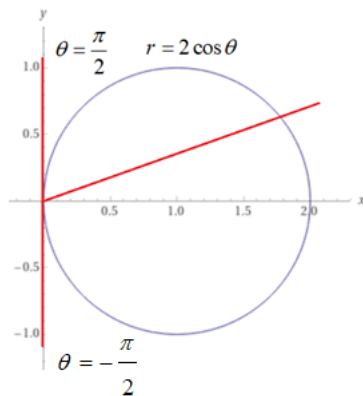
Thus, $G : 0 \leq r \leq \sec \theta , 0 \leq \theta \leq \frac{\pi}{3}$.

7. The region enclosed by the circle $x^2 + y^2 = 2x$.

極座標與 Cartesian 座標 : $x = r \cos \theta$, $y = r \sin \theta$, $x^2 + y^2 = r^2$

$$x^2 - 2x + y^2 = 0 \Rightarrow (x^2 - 2x + 1) + y^2 = 1 \Rightarrow (x-1)^2 + y^2 = 1 \quad \text{圓心}(1,0) , \text{半徑 } 1$$

$$x^2 + y^2 = 2x \Rightarrow r^2 = 2r \cos \theta \Rightarrow r = 2 \cos \theta \Rightarrow 0 \leq r \leq 2 \cos \theta$$



積分區域

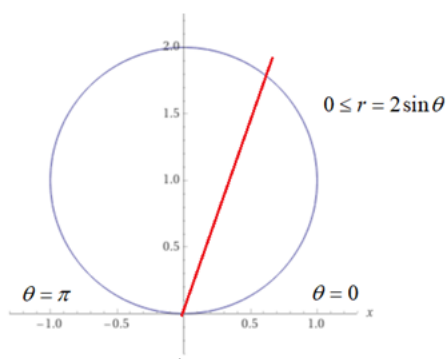
$$G: 0 \leq r \leq 2 \cos \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}.$$

8. The region enclosed by the semicircle $x^2 + y^2 = 2y$, $y \geq 0$.

極座標與 Cartesian 座標: $x = r \cos \theta$, $y = r \sin \theta$, $x^2 + y^2 = r^2$

$$x^2 + y^2 - 2y = 0 \Rightarrow x^2 + (y^2 - 2y + 1) = 1 \Rightarrow x^2 + (y-1)^2 = 1 \quad \text{圓心}(0,1) , \text{半徑 } 1$$

$$x^2 + y^2 = 2y \Rightarrow r^2 = 2r \sin \theta \Rightarrow r = 2 \sin \theta \Rightarrow 0 \leq r \leq 2 \sin \theta$$



積分區域

$$G: 0 \leq r \leq 2 \sin \theta, \quad 0 \leq \theta \leq \pi.$$

Evaluating Polar Integrals

In Exercises 9–22, change the Cartesian integral into an equivalent polar integral. Then evaluate the polar integral.

10：化為極座標積分計算積分 $I = \int_0^1 \int_0^{\sqrt{1-y^2}} (x^2 + y^2) dx dy$

解：

從積分上下限知道積分區域 $R: 0 \leq y \leq 1, 0 \leq x \leq \sqrt{1-y^2}$

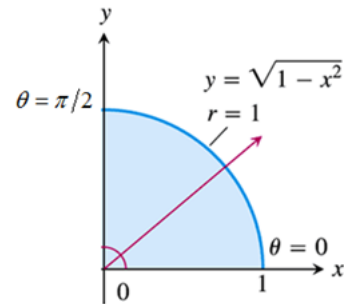
$$x = \sqrt{1-y^2} \Rightarrow x^2 = 1-y^2 \Rightarrow x^2 + y^2 = 1$$

圓心 $(0,0)$ ，半徑 1 ， R 是第一象限內單位圓圍成的區域。

出現 $f(x,y) = x^2 + y^2$ 及積分區域與圓有關

$$x = r \cos \theta, \quad y = r \sin \theta, \quad x^2 + y^2 = r^2$$

$$G: 0 \leq r \leq 1, \quad 0 \leq \theta \leq \pi/2$$



$$\text{化成極座標積分 } \iint_R f(x,y) dA = \iint_G f(r \cos \theta, r \sin \theta) \underbrace{r dr d\theta}_{dA}$$

$$I = \int_0^{\pi/2} \int_0^1 r^2 \cdot r dr d\theta = \int_0^{\pi/2} \int_0^1 r^3 dr d\theta = \int_0^{\pi/2} \frac{1}{4} r^4 \Big|_0^1 d\theta = \int_0^{\pi/2} \frac{1}{4} d\theta = \frac{1}{4} \theta \Big|_0^{\pi/2} = \frac{\pi}{8}$$

17：化為極座標積分計算積分 $I = \int_{-1}^0 \int_{-\sqrt{1-x^2}}^0 \frac{2}{1+\sqrt{x^2+y^2}} dy dx$ 。

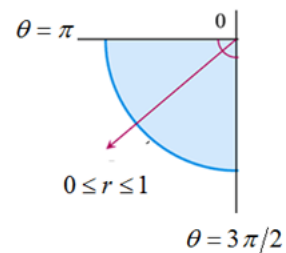
解：從積分上下限知道積分區域 $R: -1 \leq x \leq 0, -\sqrt{1-x^2} \leq y \leq 0$

$$y = -\sqrt{1-x^2} \Rightarrow y^2 = 1-x^2 \Rightarrow x^2 + y^2 = 1$$

圓心 $(0,0)$ ，半徑 1 ， R 是第三象限內單位圓圍成的區域

$$x = r \cos \theta, \quad y = r \sin \theta, \quad x^2 + y^2 = r^2$$

$$G: 0 \leq r \leq 1, \quad \pi \leq \theta \leq 3\pi/2$$



$$\text{化成極座標積分 } \iint_R f(x,y) dA = \iint_G f(r \cos \theta, r \sin \theta) \underbrace{r dr d\theta}_{dA}$$

$$I = \int_{\pi}^{3\pi/2} \int_0^1 \frac{2}{1+r} \cdot r dr d\theta = 2 \int_{\pi}^{3\pi/2} \left(\int_0^1 \frac{r}{1+r} dr \right) d\theta = 2 \int_{\pi}^{3\pi/2} \left(\int_0^1 \left(1 - \frac{1}{1+r}\right) dr \right) d\theta$$

$$= 2 \int_{\pi}^{3\pi/2} [r - \ln(1+r)]_0^1 d\theta = 2 \int_{\pi}^{3\pi/2} (1 - \ln 2) d\theta = 2(1 - \ln 2) \theta \Big|_{\pi}^{3\pi/2} = (1 - \ln 2)\pi$$

22：化為極座標積分計算積分 $I = \int_1^2 \int_0^{\sqrt{2x-x^2}} \frac{1}{(x^2+y^2)^2} dy dx$ 。

解：

從積分上下限知道積分區域 $R: 1 \leq x \leq 2, 0 \leq y \leq \sqrt{2x-x^2}$

畫出積分區域圖形

$$y = \sqrt{2x-x^2} \Rightarrow y^2 = 2x-x^2 \Rightarrow (x^2-2x+1)+y^2=1 \Rightarrow (x-1)^2+y^2=1$$

圓心(1,0)，半徑 1

$$y = \sqrt{2x-x^2} \Rightarrow y^2 = 2x-x^2 \Rightarrow x^2+y^2=2x$$

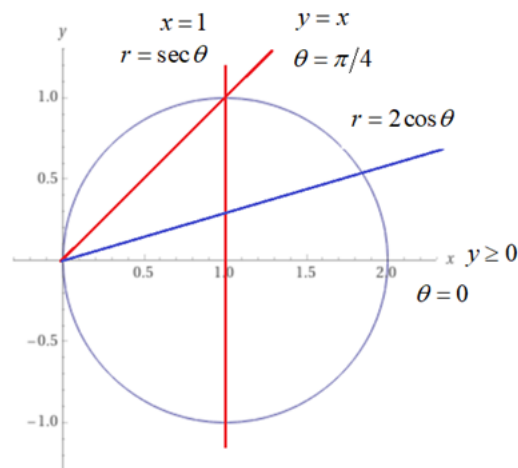
$$\Rightarrow r^2 = 2r \cos \theta \Rightarrow r = 2 \cos \theta \Rightarrow 0 \leq r \leq 2 \cos \theta$$

$$R: 1 \leq x \leq 2, 0 \leq y \leq \sqrt{2x-x^2}$$

$$x=1 \Rightarrow r \cos \theta = 1 \Rightarrow r = \frac{1}{\cos \theta} \Rightarrow r = \sec \theta$$

the ray enters at the vertical line $r = \sec \theta$,
and leaves at the circle $r = 2 \cos \theta$.

i.e., $\sec \theta \leq r \leq 2 \cos \theta$.



$$y \geq 0 \Rightarrow$$

the angle rotates counterclockwise from $\theta = 0$ to $\theta = \pi/4$.

極座標積分區域 $G: \sec \theta \leq r \leq 2 \cos \theta, 0 \leq \theta \leq \pi/4$

$$\text{化成極座標積分 } \iint_R f(x,y) dA = \iint_G f(r \cos \theta, r \sin \theta) \underbrace{r dr d\theta}_{dA}$$

$$I = \int_0^{\pi/4} \int_{\sec \theta}^{2 \cos \theta} \frac{1}{r^4} \cdot r dr d\theta = \int_0^{\pi/4} \int_{\sec \theta}^{2 \cos \theta} \frac{1}{r^3} dr d\theta = \int_0^{\pi/4} \left(\frac{-1}{2r^2} \right)_{\sec \theta}^{2 \cos \theta} d\theta = \int_0^{\pi/4} \left(\frac{-1}{8 \cos^2 \theta} - \frac{-1}{2 \sec^2 \theta} \right) d\theta$$

$$= \int_0^{\pi/4} \left(\frac{1}{2} \cos^2 \theta - \frac{1}{8} \sec^2 \theta \right) d\theta = \int_0^{\pi/4} \left(\frac{1}{2} \frac{1 + \cos 2\theta}{2} - \frac{1}{8} \sec^2 \theta \right) d\theta$$

$$= \left(\frac{1}{4} \theta + \frac{1}{8} \sin 2\theta - \frac{1}{8} \tan \theta \right) \Big|_0^{\pi/4} = \left(\frac{1}{4} \cdot \frac{\pi}{4} + \frac{1}{8} \cdot 1 - \frac{1}{8} \cdot 1 \right) = \frac{\pi}{16}$$

習題 15.8

練習作業 Homework : 2,7,13

代換 u, v 的選取，一樣與積分函數有關，除此之外也與積分區域有關。

習 2a：以變換 $u = x + 2y$, $v = x - y$ ，計算 Jacobian $J(u, v) = \frac{\partial(x, y)}{\partial(u, v)}$

解：

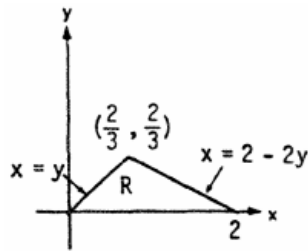
$$\begin{cases} u = x + 2y \cdots (1) \\ v = x - y \cdots (2) \end{cases}$$

$$(1) - (2) \Rightarrow u - v = 3y \Rightarrow y = \frac{u - v}{3} \quad \text{帶入 (2) 式 } x = y + v, \quad x = \frac{u}{3} + \frac{2v}{3}$$

$$\text{Jacobian } J(u, v) = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} 1/3 & 2/3 \\ 1/3 & -1/3 \end{vmatrix} = -\frac{1}{9} - \frac{2}{9} = -\frac{1}{3}$$

習 2b：若變換 $u = x + 2y$, $v = x - y$ ， R 是 xy 平面上由直線 $y = 0$, $y = x$ 及直線 $x + 2y = 2$ 所圍成的三角形區域，求 R 的像(image) G 及在 uv 平面上繪出 G 。

解：



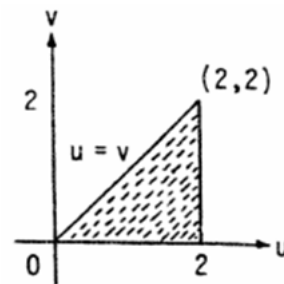
以 $x = \frac{u}{3} + \frac{2v}{3}$ 及 $y = \frac{u - v}{3}$ 帶入 R 的邊界方程式得區域 G 對應的邊界方程式

$$y = 0 \Rightarrow \frac{u - v}{3} = 0 \Rightarrow u = v$$

$$y = x \Rightarrow \frac{u - v}{3} = \frac{u}{3} + \frac{2v}{3} \Rightarrow v = 0$$

$$x + 2y = 2 \Rightarrow \frac{u}{3} + \frac{2v}{3} + 2\left(\frac{u - v}{3}\right) = 2 \Rightarrow u = 2$$

區域 G 是 uv 平面上由直線 $u = 2$, $v = 0$ 及直線 $u = v$ 所圍成的三角形區域



習 7：以變換 $u = 3x + 2y$, $v = x + 4y$ ，計算 $I = \iint_R (3x^2 + 14xy + 8y^2) dx dy$ 。

區域 R 是由直線 $y = (-3/2)x + 1$, $y = (-3/2)x + 3$, $y = (-1/4)x$ 及 $y = (-1/4)x + 1$ 所圍成
解：

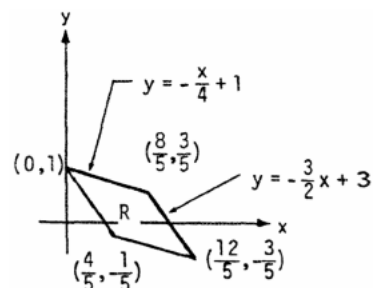
$3x^2 + 14xy + 8y^2 = (3x + 2y)(x + 4y)$ ，故代換 u, v 的選取，與積分函數有關，瞭解？

步驟 1：將 x, y 以 u, v 表示

$$u = 3x + 2y \cdots (1), \quad v = x + 4y \cdots (2)$$

$$(1) \times (-2) + (2) \Rightarrow -2u + v = -5x \Rightarrow x = \frac{2}{5}u + \frac{1}{5}v$$

$$(1) + (2)(-3) \Rightarrow u - 3v = -10y \Rightarrow y = \frac{1}{10}u + \frac{3}{10}v$$



步驟 2：從 R 的邊界方程式得區域 G 對應的邊界方程式

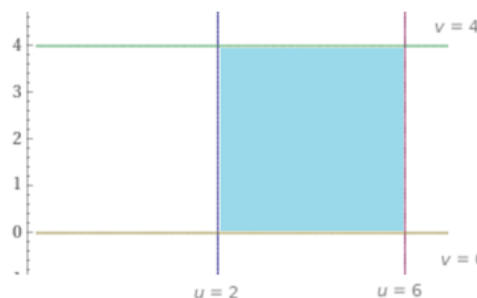
這題有比較簡單的技巧，因為 $u = 3x + 2y$, $v = x + 4y$

$$y = -\frac{3}{2}x + 1 \Rightarrow 3x + 2y = 2 \Rightarrow u = 2$$

$$y = -\frac{3}{2}x + 3 \Rightarrow 3x + 2y = 6 \Rightarrow u = 6$$

$$y = -\frac{1}{4}x \Rightarrow x + 4y = 0 \Rightarrow v = 0$$

$$y = -\frac{1}{4}x + 1 \Rightarrow x + 4y = 4 \Rightarrow v = 4$$



區域 G 是 uv 平面上由直線 $u = 2$, $u = 6$ 及 $v = 0$, $v = 4$ 所圍成的矩形區域

步驟 3：計算 Jacobian， $J(u, v) = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} \frac{2}{5} & \frac{1}{5} \\ \frac{1}{10} & \frac{3}{10} \end{vmatrix} = \frac{6}{50} - \frac{1}{50} = \frac{1}{10}$

步驟 4：積分，

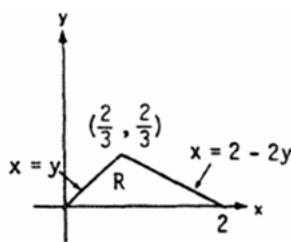
$$\begin{aligned} I &= \iint_R (3x^2 + 14xy + 8y^2) dx dy = \iint_R (3x + 2y)(x + 4y) dx dy = \iint_G uv \left(\frac{1}{10}\right) du dv \\ &= \int_0^4 \int_2^6 uv \left(\frac{1}{10}\right) du dv = \frac{1}{10} \int_2^6 u du \cdot \int_0^4 v dv = \frac{1}{10} \left(\frac{1}{2}u^2\right)_2^6 \cdot \left(\frac{1}{2}v^2\right)_0^4 = \frac{1}{10} \cdot \frac{1}{2} (32) \cdot \frac{1}{2} (16) = \frac{64}{5} \end{aligned}$$

習 13：計算 $I = \int_0^{2/3} \int_y^{2-2y} (x+2y)e^{y-x} dx dy$

解：

步驟 1：xy 平面的積分區域 $R: y \leq x \leq 2-2y, 0 \leq y \leq 2/3$

令等式成立，得 $x = y, x = 2 - 2y$ ，即 $y = x, x + 2y = 2$ 作出區域 R 的圖形



步驟 2：代換選取 $\begin{cases} u = x + 2y \cdots (1) \\ v = x - y \cdots (2) \end{cases}$ ，與積分函數 $(x+2y)e^{y-x}$ 有關，瞭解？

將 x, y 以 u, v 表示

$$(1) - (2) \Rightarrow u - v = 3y \Rightarrow y = \frac{u}{3} - \frac{v}{3} \text{ 帶入 (2) 式 } x = y + v, x = \frac{u}{3} + \frac{2v}{3}$$

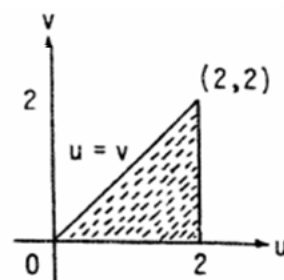
步驟 3：以 $x = \frac{u}{3} + \frac{2v}{3}$ 及 $y = \frac{u}{3} - \frac{v}{3}$ 帶入 R 的邊界方程式得區域 G 對應的邊界方程式

$$y = 0 \Rightarrow \frac{u}{3} - \frac{v}{3} = 0 \Rightarrow u = v$$

$$y = x \Rightarrow \frac{u}{3} - \frac{v}{3} = \frac{u}{3} + \frac{2v}{3} \Rightarrow v = 0$$

$$x + 2y = 2 \Rightarrow \frac{u}{3} + \frac{2v}{3} + 2(\frac{u}{3} - \frac{v}{3}) = 2 \Rightarrow u = 2$$

區域 G 是 uv 平面上由直線 $u = 2, v = 0$ 及直線 $u = v$ 所圍成的三角形區域



步驟 4：計算 Jacobian $J(u, v) = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} 1/3 & 2/3 \\ 1/3 & -1/3 \end{vmatrix} = -\frac{1}{9} - \frac{2}{9} = -\frac{1}{3}$

步驟 5：積分，

$$I = \int_0^{2/3} \int_y^{2-2y} (x+2y)e^{y-x} dx dy = \int_0^2 \int_0^u u e^{-v} (\frac{1}{3}) dv du \quad \text{注意 } |J(u, v)| = \frac{1}{3}$$

$$= \frac{1}{3} \int_0^2 u (-e^{-v})_0^u du = \frac{1}{3} \int_0^2 u (1 - e^{-u}) du = \frac{1}{3} \int_0^2 u du - \frac{1}{3} \int_0^2 u e^{-u} du$$

$$= \frac{1}{3} (\frac{1}{2} u^2)_0^2 + \frac{1}{3} (u e^{-u} + e^{-u})_0^2 = \frac{2}{3} + \frac{1}{3} (3e^{-2} - 1) = \frac{1}{3} + e^{-2}$$